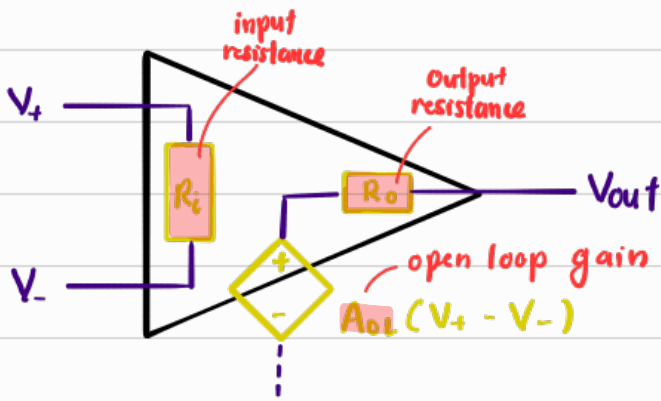


Simple non-ideal op-amp model

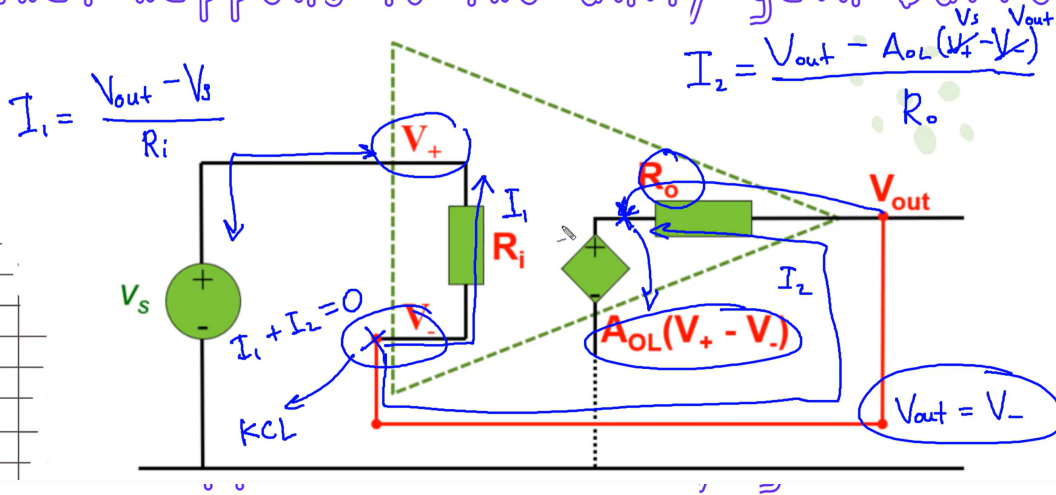


Since this is non-ideal,

- 1) $V_+ = V_{in}$ [Negative Feedback]
- 2) $i_{input} = 0$

DO NOT HOLD

What happens to the unity gain buffer?



$$i_1 + i_2 = 0$$

$$i_1 = (V_{out} - v_s)/R_i$$

$$i_2 = (V_{out} - A_{OL}(V_s - V_{out}))/R_o$$

$$0 = \frac{V_{out} - v_s}{R_i} + \frac{V_{out} - A_{OL}(v_s - V_{out})}{R_o}$$

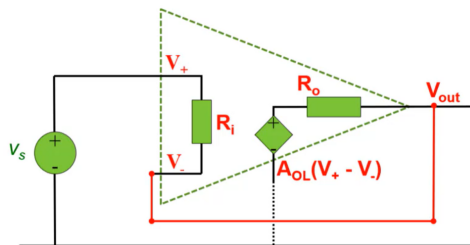
$$= R_o(V_{out} - v_s) + R_i V_{out} + R_i A_{OL}(V_{out} - v_s)$$

$$= (V_{out} - v_s)(R_o + R_i A_{OL}) + R_i V_{out}$$

$$R_i = -\left(\frac{V_{out} - v_s}{V_{out}}\right)(R_o + R_i A_{OL})$$

$$-\left(1 - \frac{v_s}{V_{out}}\right) = \frac{R_i}{R_o + R_i A_{OL}} \Rightarrow$$

$$\frac{v_s}{V_{out}} = 1 + \frac{R_i}{R_o + R_i A_{OL}}$$



Recall:

For ideal $\frac{V_{out}}{V_s} = G = 1$

$$\frac{V_{out}}{v_s} = \frac{1}{1 + \frac{R_i}{R_o + R_i A_{OL}}}$$

Open-loop gain

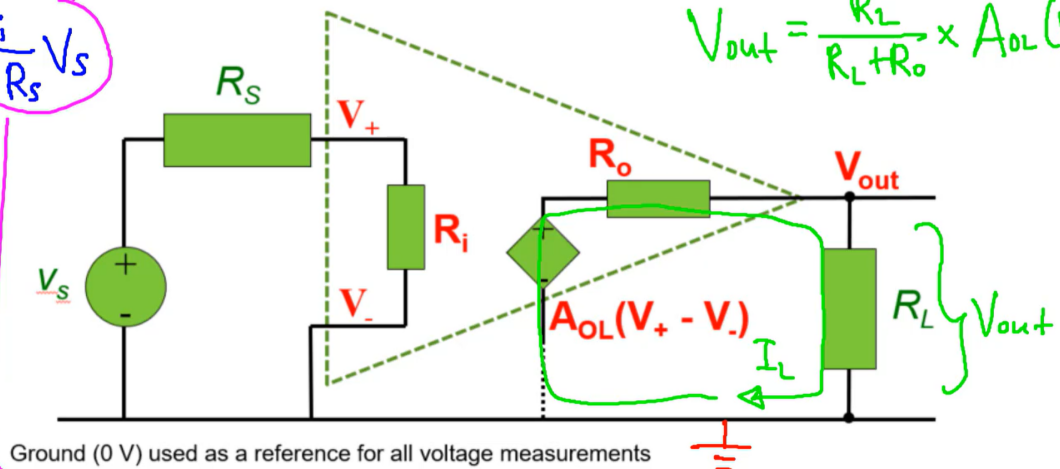
ideal; $V_{out} = A_{OL} (V_+ - V_-)$

But the "feedback" is connected to GND

NOTE: This circuit is **NOT** in negative feedback.

$$V_{R_i} = V_+ - V_- = \frac{R_i}{R_i + R_s} V_s$$

$$V_{out} = \frac{R_L}{R_L + R_o} \times A_{OL} (V_+ - V_-)$$



Open-loop gain

After the analysis, we get:

$$\frac{V_{out}}{v_s} = A_{OL} \left(\frac{R_i}{R_i + R_s} \right) \left(\frac{R_L}{R_o + R_L} \right)$$

If it is an ideal op-amp, $R_i \rightarrow \infty$, $R_o \rightarrow 0$

$$\frac{V_{out}}{v_s} = A_{OL}$$

Characteristics of real op-amp

Characteristic	Physical Definition	Practical Effect
Input Bias Current	The average microscopic DC current required by the input transistors to stay active.	Can cause unwanted voltage drops if connected to high-resistance circuits.
Input Offset Current	The difference in the bias current required by the two unmatched input pins.	Creates small error voltages in the circuit.
Input Offset Voltage	The tiny phantom voltage difference caused by unmatched internal transistors.	Causes the output to be non-zero even when the inputs are exactly equal.
CMRR	The ability of the internal circuitry to physically block voltages common to both inputs.	Higher CMRR = Better rejection of background electrical noise.
Output Swing	The absolute maximum and minimum voltage the output pin can physically reach.	Always slightly less than the power supply (V_{DD} and V_{SS}) due to internal transistor voltage drops.
Slew Rate	The maximum physical speed at which the output voltage can change ($V/\mu s$).	High-frequency or rapidly changing signals will distort if they exceed this speed limit.